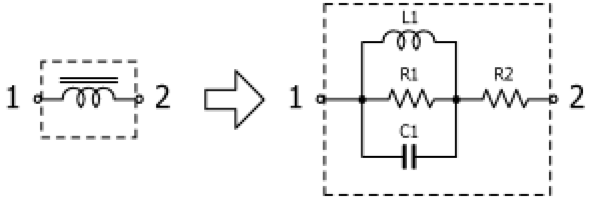
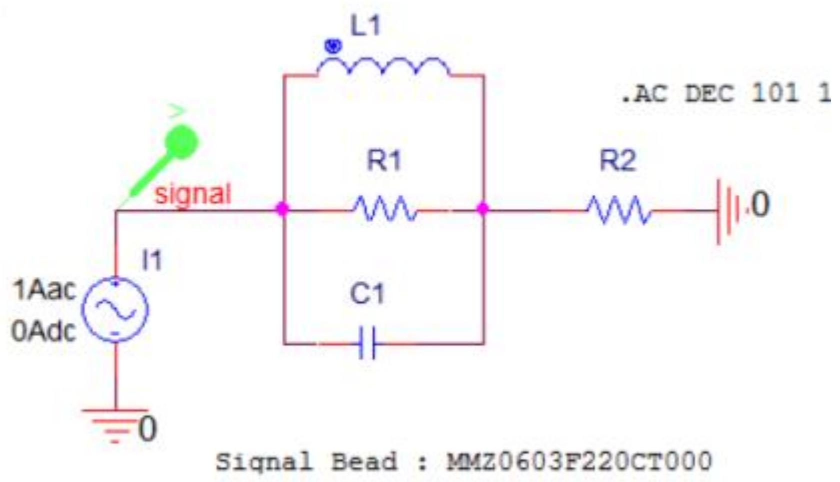
Signal/Power Beads

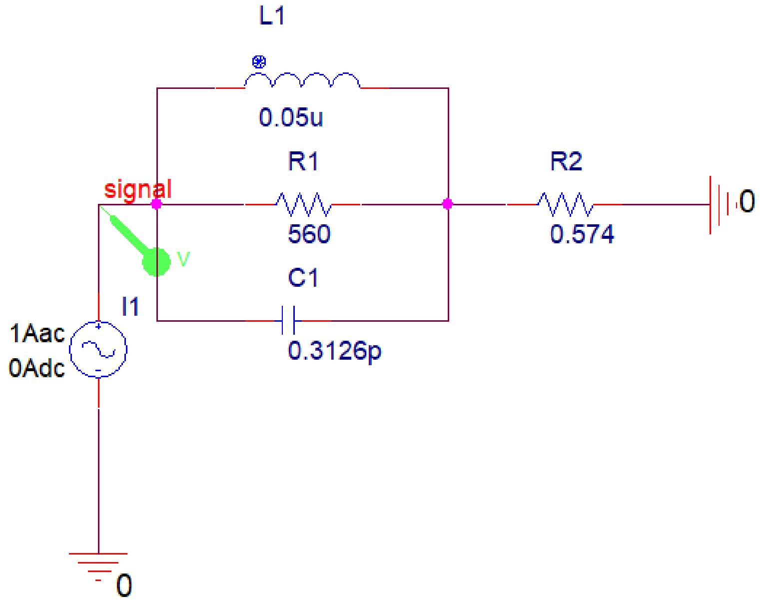
엄민호

DUT(Device Under Test) 소자 : Signal Beads (MMZ0603F220CT000)

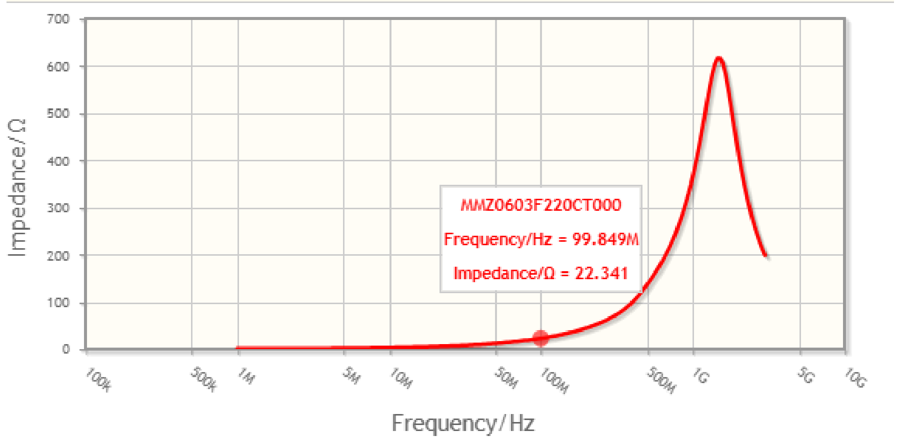


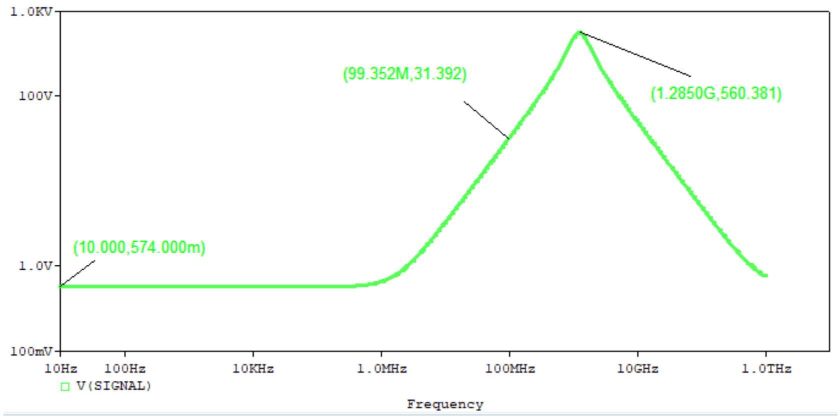
Part No.	R1[ohm]	L1[uH]	C1[pF]	R2[ohm]
MMZ0603F220CT000	560	0.050	0.3126	0.574





주파수에 따른 Signal Beads의 impedance





Impedance at 100MHz 22Ω ±25%

100Mhz에서 Impedance 값
Data sheet : 22.3 Ω
Or-CAD Simulation : 31.4 Ω

100Mhz 측정이유 : datasheet
오차율 : (|31.2-22.3| / 31.2)*100 =40.8%
오차이유 : datasheet 에서 정격전류 200mA

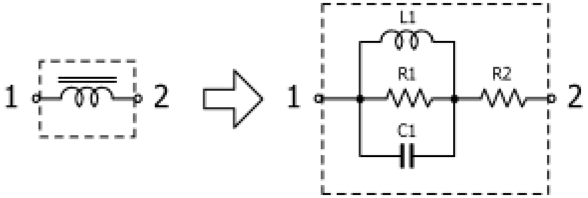
Peak 지점 Impedance 값
Data sheet : 616 Ω
Or-CAD Simulation : 560 Ω

L,C 병렬공진 저항 무한대
Impedance = R1+R2 = 560.574 Ω

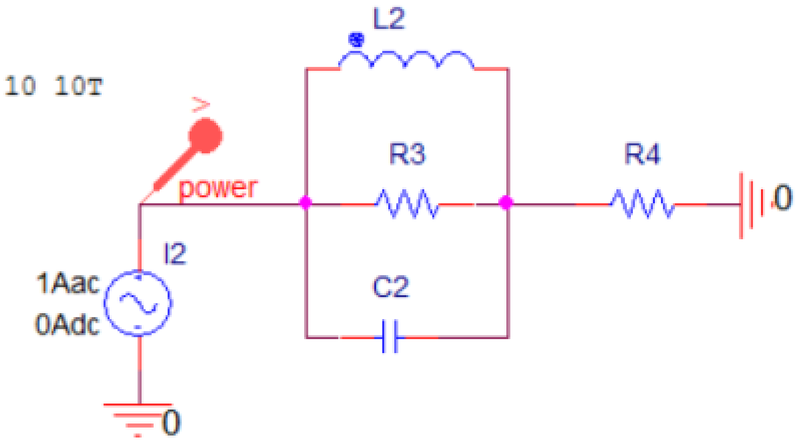
Low Frequency 에서 Impedance 값:
Data sheet : 574m Ω (at 1Mhz)
Or-CAD Simulation :657m Ω (at 10hz)

Low Frequency에서 Inductor short
Impedance = R2 = 574m Ω

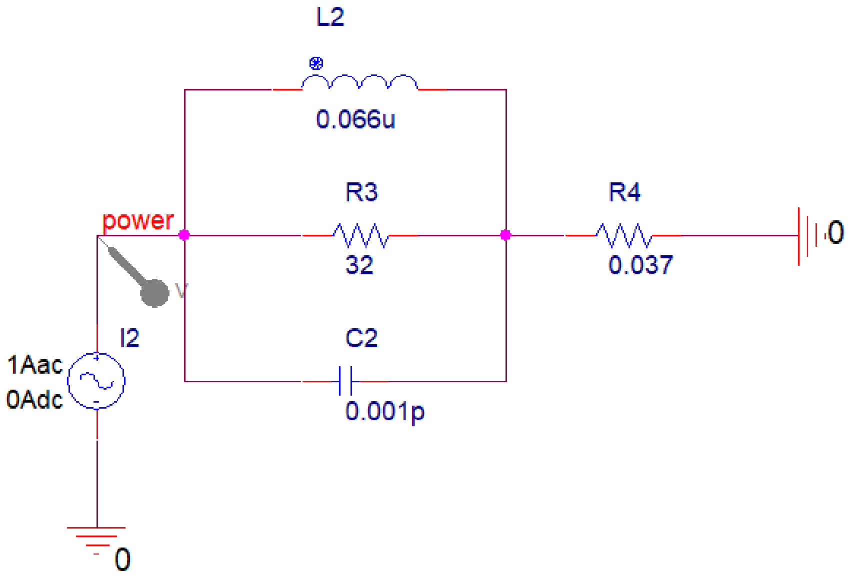
DUT(Device Under Test) 소자 : Power Beads (MPZ0603S220CT000)



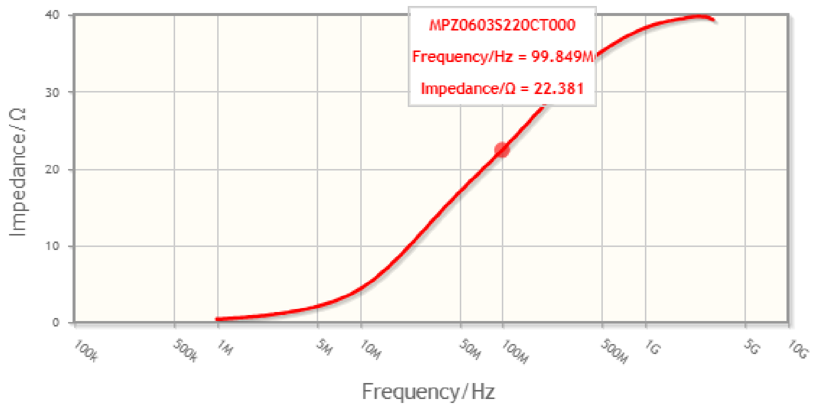
Part No.	R1[ohm]	L1[uH]	C1[pF]	R2[ohm]
MPZ0603S220CT000	32	0.066	0.0010	0.037

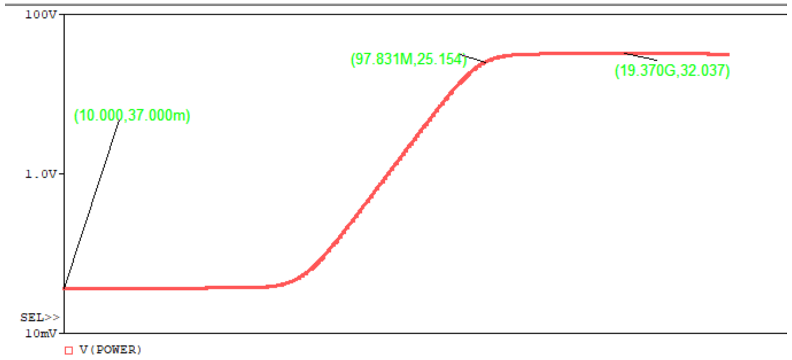


Power Bead : MPZ0603S220CT000



주파수에 따른 Power Beads의 impedance





Impedance at 100MHz	22Ω ±25%

100Mhz에서 Impedance 값
Data sheet : 22.4 Ω
Or-CAD Simulation : 25.2 Ω

Datasheet에서 100Mhz 에서 Impedance
오차율 : (|22.4-25.2| / 22.4) * 100 = 12.5%

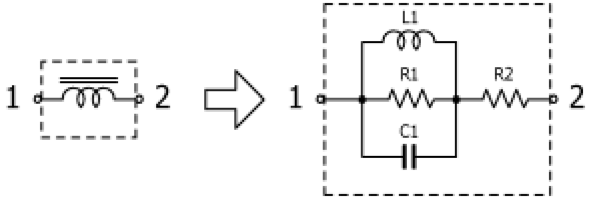
Peak 지점 Impedance 값
Data sheet : 32 Ω
Or-CAD Simulation : 39.7 Ω

Peak 지점 LC병렬공진 Impedance 무한대
Impedance = R3 + R4 = 32.037 Ω

Low Frequency에서 Impedance 값:
Data sheet : 37m Ω (at 10hz)
Or-CAD Simulation :413m Ω (at 1Mhz)

Low Frequency에서 Inductor short
Impedance = R4 = 37m Ω

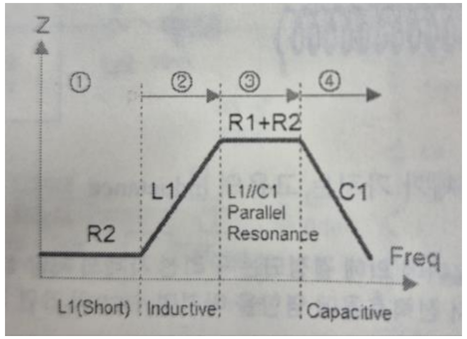
Signal VS Power Beads

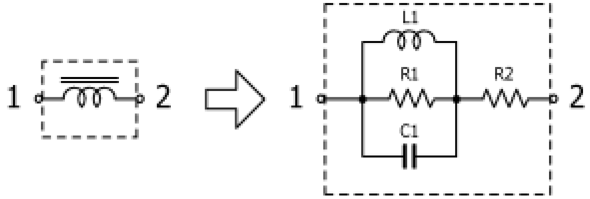


Part No.	R1[ohm]	L1[uH]	C1[pF]	R2[ohm]
MMZ0603F220CT000	560	0.050	0.3126	0.574

<Signal Beads>

- 특정 주파수 영역에서 임피던스를 가져야 함
- SRF에서 noise 제거↑ -> R1, R2 ↑
- 고주파신호도 통과필요 C1 ↑





Part No.	R1[ohm]	L1[uH]	C1[pF]	R2[ohm]
MPZ0603S220CT000	32	0.066	0.0010	0.037

<Power Beads>

- 전원공급을 위해 큰 전류 특성을 가져야함
- 저항 Impedance(R2)가 작아야함
- 전원 noise 가 광대역에 걸쳐 있어서 고주파에서도 광대역 Impedance -> 낮은 C1 값